

Requirements

Task 1

1. **Do not just put screenshots** of your code in the report. Please write down your explanation.
2. Plot your result like Fig. 1 and put the figure in the report.

Task 2

1. Design the function `myChebyPol()` **without using built-in functions**.
2. You have to write another .m file called `myChebyPol`, and then call the function in your main .m file. **Do not just put screenshots** of code in the report. Please write down your explanation.
3. Plot your result like Fig. 2 and put the figure in the report.

Discussion

1. Discuss the results of magnitude response **with different ω_s , ω_p , and L** .
2. Compare this approach and FIR design by windowing. **Which performs better?**
3. Discuss if there is a better way of choosing $x_1 < x_2 < \dots < x_{L+2}$. Please describe with detailed explanation.